

Referee report on NST-2025-0295

Let me start with the principal conclusion: the paper is definitely worth of publication as a regular article in NST after some improvement of the presentation.

The proposed quasi-frozen spin (QFS) approach is an excellent illustration to Norman Ramsay's dictum that frequency is the only observable which can be measured with really high precision. A crucial virtue of QFS approach is that at a price of modest modifications it can be implemented at existing intermediate energy storage rings.

My general impression is that English of the manuscript needs improvement. I dared to make some suggestions but proofreading by people fluent in English would be most welcome.

Here follow more specific comments on the manuscript:

Abstract, line 4 and elsewhere: energy level $\rightarrow$ energy range

Below the line numbering refers to the main text:

Line 2:

investigation of $\rightarrow$ search for

Line 7:

Complement Ref. [1] with more recent reviews

Timothy Chupp, Peter Fierlinger, Michael Ramsey-Musolf, Jaideep Singh, **Electric dipole moments of atoms, molecules, nuclei, and particles**, Rev.Mod.Phys. 91 (2019) 1, 015001

F. Abusaif et al (CPEDM Collaboration), *Storage ring to search for electric dipole moments of charged particles: Feasibility study*, (CPEDM Collaboration), *Storage ring to search for electric dipole moments of charged particles: Feasibility study*, CERN Yellow Reports: Monographs, Vol.3, CERN-2021-003 (CERN, Geneva, 2021) 246 pp.

S.N. Vergeles, N.N. Nikolaev, Yu.N. Obukhov, A.Ya. Silenko, O.V. Teryaev, *General relativity effects in precision spin experimental tests of fundamental symmetries*, Phys. Usp., 66(2), 109-147 (2023)

Two last references give an extensive discussion of different aspects of searches for the EDM of charged particles and axions in storage ring experiments.

Line 14:

segregated $\rightarrow$ separated

Lines 28-29:

its dynamics evaluated in external electromagnetic fields using

→ its dynamics in external electromagnetic fields is described by

Lines 39-41:

I'm puzzled by the statement

The simplest approach, initially proposed at BNL [3], involves fully compensating the magnetic field with an electric field aligned along the momentum.

Note that permanent electric field along the particle momentum would either accelerate or decelerate the beam which is unacceptable.

Authors must consult Anastassopoulos, V. et al, 2016. A storage ring experiment to detect a proton electric dipole moment. *Review of Scientific Instruments*, 87 (2016) 11, 115116; also available as an e-print [1502.04317](#) [physics.acc-ph] and cite this reference instead of the current one.

Most likely, authors wanted to say “The simplest approach, initially proposed at 40 BNL [3], involves **full elimination** of the magnetic field **in favor of confinement of particles by the radial** electric field.” ?

Line 55:

unaltered → unperturbed

Line 65:

violate → perturb

Line 72:

compensator → compensation

Lines 118-119:

Maximum achievable electric field values 10-13 V/m [5]

→ Maximum **attainable** electric fields **are** 10-13 V/m [5],

Lines 127-129:

The maximum analyzing capability of the polarimeter is reached at a beam energy of 270 MeV [6]

→ The maximum **figure of merit** of the polarimeter is reached at a proton beam energy of 270 MeV [6] **and the same is true for deuterons [**]**

Here [**] stands for

F. Müller et al, *Measurement of deuteron carbon vector analyzing powers in the kinetic energy range 170-380 MeV*, Eur. Phys J. A 58(8), 211 (2020)

Lines 142-147;

Improve your English:

For protons this effect is so important as spin rotation per arc should be limited $\gamma G \cdot \Phi_{\text{arc}} p < \pi/2$ to save an ability for EDM accumulation for longitudinally polarized beam. And usage of periodic structures can save the idea of measuring proton EDM with QFS lattice, should be at least $N = 8$ and greater.

→ For protons this effect is so important that a care must be taken to assure the spin rotation per arc $\gamma G \cdot \Phi_{\text{arc}} p < \pi/2$ to preserve accumulation of the EDM effect for longitudinally polarized beam, so that QSF lattices only with $N=8$ or larger give a chance to measure the proton EDM.

Lines 153-155:

For various particle and structure periodicity it states at Table 1. Maximum periodicity considered as $N = 16$, due to lattice design difficulties.

→ For various particle and structure periodicity it is cited in Table 1. Maximum periodicity considered is $N = 16$, due to lattice design limitations.

Which accelerator do authors keep in mind?

Lines 155-156:

The sentence “Shown that for deuterons QFS lattice close to FS” is incomprehensible.

Evidently, the intended message was that for stored deuterons lattices for the quasifrozen-spin and frozen-spin are close to each other ?

Lines 156-157: For protons, 16-periodic structure can provide a real opportunity to practice the methodology

→ For protons, 16-periodic structure offers a real opportunity of practical tests of the QFS approach.

Table 1:

specify the beam energy/momentum. Still better if Table 2 would precede Table 1

Lines 181-182:

What is a message in “which is useful for methodological refinement but insufficient for statistical validation (Table 2).” ?

Fig. 1:

Indicate the magic energy for protons.

Line 191:

Clarify the point “not violate spin by EDM”

Lines 206-208

Overall, the Wien filter’s ability to control particle spin without perturbing orbital paths makes it an invaluable component in advanced accelerator experiments.

This point is very well taken. Note that A. Silenko and, from a somewhat different perspective, N. Nikolaev suggested Wien filters to convert spins in storage rings into broadband axion antennas:

[A.J. Silenko](#) , ([Dubna, JINR](#) and [Lanzhou, Inst. Modern Phys.](#) and [Minsk, Inst. Nucl. Problems](#)) [Relativistic spin dynamics conditioned by dark matter axions](#), Eur.Phys.J.C 82 (2022) 10, 856

N.N. Nikolaev, *Spin of protons in NICA and PTR storage rings as an axion antenna*, JETP Letters, 115(11), 639-643 (2022)

Mentioning axions would have been up to the point and enliven the paper.